

Chapter 83 — Tumours of the Urethra

BENIGN URETHRAL TUMOURS

- Rare, documented almost entirely in small case reports; commonest are **leiomyoma**, **haemangioma** and **fibroepithelial polyp (FEP)**.
- Female urethral prolapse and urethral caruncle are benign mimics, not true tumours.

COMPARISON: THE THREE BENIGN URETHRAL TUMOURS

Feature	Leiomyoma	Haemangioma	Fibroepithelial polyp
Origin	Urethral/paraurethral smooth muscle	Angioblastic cells	Mesodermal
Commonest in	Reproductive-aged females	Males ; no age predilection; solitary or multiple	Males, usually children
Site	Anywhere; mostly posterior wall	Anywhere along the length	Bulbar and posterior urethra
Symptoms	Obstructive voiding, dyspareunia, haematuria; often incidental	Intermittent haematuria, bloody discharge ; painless	Obstructive symptoms, haematuria, dysuria, rarely retention
Endoscopy	Suburothelial, no ulceration ; firm smooth anterior vaginal wall mass	Bluish sessile lesion with varicosities ; cavernous histology	Smooth pink/tan tumour on a stalk ; may protrude from meatus in females
Imaging	Ultrasound hypoechoic, internally vascular ; MRI T1 isointense to muscle, T2 hypointense, enhancing	MRI high T2 , defines true extent	Smooth cylindrical filling defect on urethrography and voiding cystourethrography
Treatment	Excision curative in nearly all ; transvaginal or transurethral by size/site	Small: fulguration or laser. Extensive: open excision with reconstruction. Embolisation described	Transurethral resection, usually curative
Histology	Smooth muscle fascicles, uniform spindle nuclei, no atypia or mitoses	Cavernous haemangioma	Fibrovascular core, normal overlying urothelium
Recurrence	No metastasis; local recurrence described	Recur if incompletely ablated	Rare after complete resection

- Purpose: **each lesion is identified endoscopically before histology**, so the endoscopic signature separates them.
- Pattern: **all three are cured by complete local removal**; reported failures are completeness failures, not biological ones.
- Trap: benign does not mean non-recurrent — **leiomyoma recurs locally, haemangioma recurs after partial ablation**.
- Leiomyoma is **hormonally sensitive**: grows in pregnancy, regresses post-partum.
- Haemangioma associations: **cutaneous haemangiomas** and **Klippel-Trénaunay-Weber syndrome**. Pain implies **thrombosis**. Diagnosis by **cystoscopy and biopsy**.
- Posterior urethral haemangioma lies **between verumontanum and external sphincter** → **haematospermia and post-ejaculatory haematuria**.

- **FEP histology is mandatory to exclude urothelial papilloma.**

Concept — *three lesions of three tissue lineages converging on one clinical problem, a mass obstructing a tube; lineage predicts the endoscopic and imaging signature, and completeness of removal is the only therapeutic variable.*

MALE URETHRAL CANCER

Epidemiology, Aetiology and Presentation

- **Male primary urethral cancer (PUC) is rare: 1.3–4.3 per million males, manifesting after the sixth decade, more common in Black than White patients.**
- **Chronic urethral inflammation underlies most anterior PUC. Stricture disease is the commonest risk factor, present in up to 50%, with latency to cancer of 20–30 years** — hence sustained suspicion rather than discharge from follow-up.
- **Other risk factors: sexually transmitted infection, lichen sclerosus, urethritis, pelvic radiotherapy, trauma, instrumentation, human papillomavirus 16, arsenic.**
- **Presentation: obstructive voiding symptoms, haematuria or bloody urethral discharge, penile or perineal mass. Nearly all are symptomatic at presentation.**

Pathology

- **Male urethra ~20 cm. Anterior = bulbar + pendulous + fossa navicularis; posterior = prostatic + membranous.**
- **Epithelial sequence: posterior urothelium → anterior stratified and pseudostratified columnar → fossa navicularis stratified squamous.**
- **Glands: paired bulbourethral (Cowper's) glands into the bulbar urethra; periurethral glands of Littre along the anterior urethra.**
- **Histology overall: urothelial 50–80%, squamous cell carcinoma (SCC) 10–30%, adenocarcinoma 5–10%; rare types melanoma and sarcoma.**
- **Registry data overstate the urothelial share through misclassification and miscoding; in institutional series urothelial carcinoma is the minority histology.**
- **Site–histology pair: posterior cancers mostly urothelial; up to 80% of anterior cancers are SCC. Adenocarcinoma relatively more common in the bulbar urethra.**
- **Spread: direct extension into corpus spongiosum and periurethral tissues; lymphatics anterior → inguinal nodes (bulbar may also drain to external iliac), posterior → pelvic nodes.**
- **Palpable inguinal nodes occur in 20–30% and almost always mean metastasis, unlike penile cancer, where they are frequently inflammatory** — so nodes in PUC are not observed through antibiotics.

TABLE 83.1 GRADING SYSTEM FOR PRIMARY URETHRAL CARCINOMA

Histology	Grade	Definition
Urothelial carcinoma	LG	Low grade
Urothelial carcinoma	HG	High grade
SCC and adenocarcinoma	Gx	Cannot be assessed
SCC and adenocarcinoma	G1 / G2 / G3	Well / moderately / poorly differentiated

Source: Hansel D, Reuter VE, Bochner B, et al. Urethra. In: Amin MB, ed. *AJCC Cancer Staging Manual*, 8th ed., 2017.

- Purpose: **grading is histology-specific, not one scale**. Trap: writing G1–G3 for a urothelial tumour — **urothelial carcinoma is two-tier only**.

Concept — *the pathology follows from one fact: the urethra changes epithelium along its length, so site predicts histology, histology predicts behaviour, and the same anterior–posterior division fixes the lymphatic basin to be treated.*

Evaluation and Staging

- **Examination under anaesthesia with cystoscopy and bimanual palpation** of external genitalia, urethra, rectum, perineum and inguinal nodes is **required** for clinical stage.
- **Tissue diagnosis by transurethral or percutaneous needle biopsy**. Urine cytology has **poor sensitivity and little utility**.
- **MRI is optimal for local extent; CT or MRI of abdomen and pelvis for metastasis; CT chest in invasive disease**.

TABLE 83.2 AMERICAN JOINT COMMITTEE ON CANCER URETHRAL CANCER TUMOUR, NODE, METASTASIS STAGING SYSTEM, 8TH EDITION

Category	Male penile urethra and female urethra	Male prostatic urethra
TX / T0	Cannot be assessed / no evidence of primary tumour	
Ta	Non-invasive papillary carcinoma	
Tis	Carcinoma in situ	Carcinoma in situ; prostatic urethra, periurethral or prostatic ducts without stromal invasion
T1	Subepithelial connective tissue	Urethral subepithelial connective tissue immediately under the urothelium
T2	Corpus spongiosum or periurethral muscle	Prostatic stroma around the ducts, by direct extension or from prostatic ducts
T3	Corpus cavernosum or anterior vagina	Periprostatic fat
T4	Other adjacent organs (e.g. bladder)	Other adjacent organs (e.g. bladder, rectum)
N1 / N2	Single / multiple regional nodes — inguinal, true pelvis (perivesical, obturator, internal [hypogastric], external iliac), or presacral	
M1	Distant metastasis	

Source: Hansel D, Reuter VE, Bochner B, et al. Urethra. In: Amin MB, ed. *AJCC Cancer Staging Manual*, 8th ed., 2017.

- Purpose: **one urethra, two T systems**, separated by the tissue surrounding each segment.
- Pattern: **T2 is always the first structure outside the urethral wall** — spongiosum distally, prostatic stroma proximally; T3 is the next compartment out.
- Traps: applying penile T categories to the prostatic urethra; **calling prostatic ductal involvement invasive when it is Tis**; reading N as size or laterality when it is **node number**.

Concept — *staging here is anatomical, not dimensional: every T category names a compartment, so knowing which compartment surrounds which segment reconstructs the system without memorising it.*

MALE ANTERIOR URETHRAL CANCER

Prognosis

- Across all males with PUC: **5-year cancer-specific survival 50–70%, overall survival 40–50%**. Survival associated with **grade, stage, location, histology**.

- Anterior tumours: **60% bulbar, 30% pendulous. Distal tumours often curable; bulbar survival as low as 20–30%.**
- Two reproducible gradients — **low stage over high stage, distal over proximal** — each a twofold to threefold separation.
- **Proximal tumours present at higher stage**, explaining much of the location effect, but **stage remains independently prognostic after controlling for location**: grade and location act largely through stage.
- **Adenocarcinoma in males may carry a more favourable prognosis.**

Treatment

- **Aggressive local control is paramount; method varies by location; interest in multimodal treatment grows with stage.**

Carcinoma of the Pendulous Urethra

- **Pendulous and fossa navicularis tumours are usually resectable.** Options escalating with size, location and stage: **transurethral resection → distal urethrectomy ± partial penectomy → total urethrectomy with perineal urethrostomy ± total penectomy.**
- **Surgical monotherapy achieves local control in distal tumours** (small case series); **low-stage fossa navicularis tumours cured by distal urethrectomy alone.**
- Penectomy gives **local control in nearly all pendulous tumours, but systemic relapse exceeds that of fossa navicularis tumours** — within the anterior urethra, more proximal origin means greater systemic threat at equal local control.

Penile preservation.

- Historically **urethral resection with radical or partial penectomy; excision with penile preservation is now an option in selected patients**, endorsed by the European Association of Urology (EAU) and National Comprehensive Cancer Network (NCCN).
- Techniques, by location and stage: **distal hypospadias with topical 5-fluorouracil (5-FU); distal urethrectomy with two-stage urethroplasty; glansectomy with skin graft; anterior urethrectomy with perineal urethrostomy.**
- **Positive margins treated by further surgery or adjuvant radiotherapy, with no local recurrence** — preservation costs no local control if margins are pursued.

Carcinoma in situ.

- **SCC in situ of the perimeatal glans extends into the distal urethra**; treated by **partial glansectomy and distal urethrectomy with reconstruction or penile urethrostomy** once the margin is negative.
- **Distal urethral carcinoma in situ (CIS) risks upstaging to invasive or node-positive disease** — **thorough staging and aggressive therapy are critical** despite the label.

Margins.

- **1 cm negative margin traditional; 5 mm gives excellent local control** and is the contemporary standard.
- **Total penectomy if invasive disease reaches the proximal penile urethra; assess proximal extent before contemplating partial urethrectomy.**
- **Local recurrence after distal resection is low, but surveillance continues.**

Inguinal lymph node management.

- **Unlike penile cancer, prophylactic inguinal lymphadenectomy has no established survival benefit in cN0 anterior PUC, and no predictors of nodal metastasis exist in cN0 disease** to select patients.

- **Largest squamous PUC analysis: only a small minority of cN0 groins prove pN1, most cN1 groins do; pN1 strongly associated with overall survival; lymphadenectomy benefited cN1 but not cN0 patients.**
- **Sentinel node biopsy** has been used to detect low-volume metastasis, proceeding to lymphadenectomy if positive. **Low-volume inguinal metastasis can be rendered disease-free by lymphadenectomy** — the rationale for aggressive nodal treatment.
- **Prophylactic lymphadenectomy for advanced pendulous PUC with cN0 groins remains controversial.** If the primary is irradiated and nodes are cN0, prophylactic nodal irradiation should be strongly considered.

Locally advanced or node-positive distal disease.

- **Surgery ± systemic therapy has variable success; cure with limited nodal disease is reported, so systemic therapy then inguinal lymphadenectomy should be considered for palpable nodes, with chemoradiotherapy an alternative.**
- **Nodal control also prevents skin breakdown, wound drainage and vascular erosion** — an indication in its own right.

Carcinoma of the Bulbar Urethra

- **Low-stage proximal anterior lesions may be treated by transurethral resection or segmental excision with end-to-end anastomosis, with repeat transurethral resection considered for staging accuracy and completeness.**
- **Few are eligible** — bulbar PUC presents at advanced stage and local control is difficult.
- **Standard for advanced bulbar cancer: radical cystoprostatectomy, pelvic lymphadenectomy, total urethrectomy and penectomy, with resection of pubic rami and adjacent urogenital diaphragm if extent demands.**
- **Bladder preservation in selected cases: urethrectomy with perineal urethrostomy, or bladder neck closure with a continent catheterisable stoma; weighed against local relapse and dissemination.**
- **Positive margins or local recurrence → chemotherapy or chemoradiotherapy.**

Technique — radical cystectomy with total urethrectomy and penectomy.

1. **Dorsal lithotomy, for simultaneous perineal access.**
 2. **Transabdominal mobilisation of the bladder, preserving endopelvic fascia and anterior pubic attachments.**
 3. **Modified lambda-shaped or inverted U-shaped perineal incision, apex mid-perineum.**
 4. **Inferior flap mobilised by dividing subcutaneous tissue and rectourethral muscle.**
 5. **Superior flap to superficial Colles fascia, then bilaterally to the adductors at the inferior pubic rami.**
 6. **Develop the ischiorectal fossae as in perineal prostatectomy; tunnel bluntly anterior to the rectum, fossa to fossa.**
 7. **Circumferential incision of skin and dartos at the penoscrotal junction; mobilise the corpora a short distance proximally to permit inferior pubectomy — not too far, or the anterior aspect of a locally advanced tumour is breached.**
 8. **Pass the penis downward through the perineal incision.**
 9. **Divide the scrotum in the midline for exposure if needed; testicles usually preserved.**
 10. **Deliver the specimen en bloc.**
- **Surgery alone: five-year overall survival for proximal tumours is about one-quarter, few disease-free** — the basis for declaring **surgery alone inadequate in advanced PUC, though more aggressive bulbar resection may fare better.**

- A negative margin may require **resection of testicles, inferior pubic rami and pelvic floor musculature**, with **myocutaneous vascularised pedicle flaps** and **orthopaedic and plastic surgical collaboration**.
- **Pelvic nodal involvement confers high risk of recurrence and death.**

Chemoradiotherapy and Radiotherapy

- **Chemoradiotherapy ± surgical consolidation is an alternative to resection in advanced disease: 45–55 Gy external beam to external genitalia, perineum and inguinal nodes with systemic 5-FU and mitomycin C (MMC).**
- **Largest male series: mostly clinical T3 or N1, majority achieved complete response including some node-positive patients. Complete response rates are similar proximally and distally — response does not follow the surgical prognostic gradient.**
- **Consolidative surgery frequently shows ypT0, confirming true pathological response, though its role after complete response is unsettled.**
- **Qualifications: about a third of complete responders recurred locally and some could not be salvaged; every responder developed a urethral stricture, often needing complex reconstruction; non-responders died of disease; surgery, not chemoradiotherapy, is recommended for low-stage distal PUC.**
- **Given the morbidity of surgery for advanced bulbar PUC, chemoradiotherapy is reasonable in selected patients and is preferred for advanced squamous cell PUC. Radiotherapy monotherapy is ineffective for bulbar PUC.**
- **Primary radiotherapy: only for early-stage anterior lesions in patients refusing surgery, using parallel opposed fields with the penis suspended vertically by a urethral catheter. Advantage penile preservation; complications skin ulceration and necrosis, urethral stricture, chronic oedema; long-term results hard to evaluate as male series are few.**

Multimodal Therapy and Systemic Chemotherapy

- **Poor outcomes after surgery alone, particularly bulbar, make surgical monotherapy inadequate in advanced PUC. Multimodal therapy = surgery plus perioperative chemotherapy ± radiotherapy.**
- **More likely delivered at high-volume academic centres; referral recommended. Uptake: only a small minority of advanced patients receive multimodal therapy, though about a third of contemporary cohorts receive perioperative chemotherapy.**
- **Cisplatin-based neoadjuvant chemotherapy or chemoradiotherapy gives complete or partial response in a substantial minority, and prognosis correlates highly with response.**
- **Chemotherapy before surgery or radiotherapy gave better survival than surgery alone or surgery plus adjuvant chemotherapy; all locally advanced or node-positive patients so treated in that series were alive at five years.**
- **Postoperative radiotherapy reaches few patients but improves survival in advanced disease; intraoperative radiotherapy may help where a negative margin is unobtainable.**
- **Medical-oncology referral cohort (majority female, SCC and adenocarcinoma predominant, almost all ≥T3): histology-directed platinum regimens gave complete or partial response in the majority, about half had surgical consolidation, and median overall survival was markedly better with surgery. Some node-positive patients were rendered disease-free by neoadjuvant chemotherapy then surgery.**
- **Neoadjuvant cisplatin-based chemotherapy is reasonable for high-risk and locally advanced PUC (EAU, NCCN); adjuvant systemic therapy is an option after surgery for locally advanced disease.**

COMPARISON: PERIOPERATIVE SYSTEMIC REGIMENS BY HISTOLOGY

Histology	Regimen	Components
Urothelial carcinoma	MVAC	Methotrexate, vinblastine, doxorubicin, cisplatin
Urothelial carcinoma	GC	Gemcitabine, cisplatin (female PUC)
Urothelial carcinoma, SCC	TIP	Paclitaxel, ifosfamide, cisplatin
Urothelial carcinoma, SCC	CGI	Cisplatin, gemcitabine, ifosfamide
Adenocarcinoma	Gem-FLP	Gemcitabine, 5-FU, leucovorin, cisplatin
Any, with radiotherapy	Radiosensitisers	Cisplatin, 5-FU, MMC, alone or combined

- Purpose: **the biopsy chooses the regimen**. Pattern: **all are cisplatin-based**; only partners change, **TIP and CGI straddle urothelial and squamous histology**, **Gem-FLP is adenocarcinoma-specific**.
- Trap: **MVAC has activity against urothelial carcinoma but not other histological types**.

Guideline Position, Recurrence and Metastatic Disease

- **Non-metastatic advanced bulbar PUC: radical surgery ± perioperative chemotherapy, chemoradiotherapy, or radiotherapy monotherapy. Clinically node-positive disease: chemoradiotherapy or chemotherapy followed by surgical consolidation.**
- **There is no high-quality comparative effectiveness research on treatment options for male PUC.**
- **At least half of advanced patients given perioperative chemotherapy recur; risk of local or systemic recurrence after resection of invasive disease approaches 50%, requiring regular urethral, inguinal and systemic surveillance. Commonest distant recurrence: lung and extrapelvic nodes. Non-metastatic local recurrence can be salvaged by surgery or radiotherapy.**
- **At presentation: 10–20% clinically node-positive, 5–20% metastatic, most often non-regional nodes, liver, lung, bone. Long-term survival is reported with cisplatin-based chemotherapy plus excision of the primary — metastatic disease is not uniformly palliative.**
- **Genomics: microsatellite instability absent in all tumours tested, but increased programmed death-ligand 1 (PD-L1) staining and tumour mutational burden, particularly in urothelial and squamous cancers; dramatic responses to checkpoint inhibition reported in programmed death-1 (PD-1)/PD-L1-positive metastatic disease.**

TABLE 83.3 SUMMARY OF 2022 NATIONAL COMPREHENSIVE CANCER NETWORK CLINICAL PRACTICE GUIDELINES ON PRIMARY URETHRAL CANCER OF THE MALE ANTERIOR AND FEMALE URETHRA

Stage and site	First-line options	Adjuvant therapy
Tis, Ta, T1	Repeat transurethral resection (TUR) ± intraurethral bacillus Calmette-Guérin (BCG) or chemotherapy	—
Male T2, pendulous	Partial urethrectomy ± partial penectomy*	Positive margin: chemoradiotherapy, further surgery, or radiotherapy
Male T2, bulbar	Urethrectomy ± radical cystectomy*	≥ pT3 or N1: chemotherapy (by histology) or chemoradiotherapy
Female T2	Radical cystectomy + urethrectomy*, or distal urethrectomy by location*, or chemoradiotherapy	—
≥T3 N0	Chemoradiotherapy ± consolidative surgery; or neoadjuvant chemotherapy with consolidative surgery or radiotherapy; or radiotherapy; or surgery alone for non-urothelial histology	—

Stage and site	First-line options	Adjuvant therapy
N1	Chemoradiotherapy (preferred for SCC); or systemic therapy ± consolidative surgery; or chemoradiotherapy ± consolidative surgery	—

*Consider neoadjuvant chemotherapy or chemoradiotherapy. Source: National Comprehensive Cancer Network Guidelines, 2022.

- Purpose: **stage and segment together fix the operation**. Pattern: **the more proximal the tumour, the more organs come out** — pendulous disease costs part of the penis, bulbar disease costs the bladder — and **every T2 option carries a neoadjuvant consideration**.
- Trap: reading the female row as cystectomy-only. **Distal urethrectomy is an equal first-line option where location permits**.

Concept — *the anterior urethra behaves as two diseases sharing a name: distal tumours are a local problem solved by adequate margins, proximal tumours are a systemic problem surgery alone cannot solve, and every recommendation expresses where on that axis the tumour sits.*

MALE POSTERIOR URETHRAL CANCER

Prognosis

- Prostatic urethral carcinoma is usually found in patients with a history of bladder cancer, not as a primary event.
- 20–40% with aggressive bladder cancer have synchronous prostatic involvement at radical cystectomy (RC); up to 39% of high-risk non-muscle-invasive bladder cancer (NMIBC) recurs in the prostate after intravesical therapy.
- Reported synchronous involvement in NMIBC varies from 3% to 30% — a methodological range, because prostatic urethral biopsies are not consistently taken at transurethral resection of bladder tumour (TURBT).
- Primary prostatic urethral carcinoma without bladder cancer is rare — under 4% of all prostate cancers; therefore urothelial carcinoma found in the prostate mandates evaluation for coexisting bladder and upper tract disease.
- May involve urothelium, ducts, acini or stroma; diagnosed by transurethral resection of the prostate (TURP); commonly presents as CIS. Staged by the same TNM system, with CT or MRI required.
- Survival: superficial or ductal involvement 67–74%; stromal involvement 36%. With stromal invasion, prognosis follows the bladder tumour stage — about 61% with T1 bladder tumours versus under 30% with muscle-invasive or locally advanced disease.

Treatment

- Management derives from bladder cancer practice. Historically prostatic involvement was called T4 and managed by RC, until it was recognised that **not all such patients do badly and many can be managed less aggressively**.
- High-grade superficial disease treated with BCG achieves prostatic response of at least 70%. Extensive TURP before BCG removes most cancerous urothelium, improves BCG exposure, may raise response rates and ensures staging — though BCG without TURP remains an option.
- With extensive endoscopic resection plus intravesical BCG, roughly **half achieve complete response but a substantial minority progress**, some requiring RC — **with no progression events within the prostate itself**.

- Ductal involvement, as a pair: **cystectomy is an indication for some and remains in guidelines**, but **TURP and BCG is effective with prognosis similar to superficial involvement**. Either way, **risk of progression and understaging mandates close follow-up with repeat prostatic biopsies**.
- **Stromal invasion** — radical cystectomy, urethrectomy and pelvic lymphadenectomy ± perioperative chemotherapy.

TABLE 83.4 SUMMARY OF 2022 NATIONAL COMPREHENSIVE CANCER NETWORK CLINICAL PRACTICE GUIDELINES ON NON-METASTATIC UROTHELIAL CARCINOMA OF THE PROSTATE IN PATIENTS WITH OR WITHOUT CONCURRENT BLADDER CANCER

Involvement	First-line treatment options	Prognostic band
Mucosal	TURP and BCG	5-year survival 67–74%
Ductal or acinar	TURP and BCG, or radical cystectomy ± urethrectomy	5-year survival 67–74%
Stromal invasion	Radical cystectomy ± urethrectomy; consider neoadjuvant chemotherapy	5-year survival 36% , governed by bladder tumour stage

Source: National Comprehensive Cancer Network Guidelines, 2022; survival bands from the chapter text.

- Purpose: one boundary — **stroma or not**. Pattern: **the first two rows are the same disease and the same treatment**; only the third changes operation and prognosis.
- Trap: **treating ductal involvement as stromal invasion** and going straight to cystectomy, when **ductal disease without stromal invasion is Tis and responds to TURP and BCG**.

KEY POINTS: MALE URETHRAL CANCER

- **Distal anterior male PUC has a better prognosis than proximal anterior PUC**, which is often associated with local invasion and distant metastasis.
- **Multimodal therapy should be considered for advanced urethral tumours**, given poor outcomes after surgery and radiation alone.
- **Benefit from prophylactic inguinal lymph node dissection has not been demonstrated in anterior urethral cancer**.
- **Primary posterior urethral carcinoma is rare, and non-invasive tumours can be effectively managed with TURP and BCG**.

Concept — *posterior urethral carcinoma is best understood not as a urethral cancer but as bladder cancer manifesting in a second compartment, which is why its staging, treatment and prognosis are imported from bladder practice and why the only question asked of the pathology is whether the stroma is breached.*

FEMALE URETHRAL CANCER

Epidemiology, Aetiology and Presentation

- **Fewer than 100 cases annually in the United States; incidence 0.5–1.5 per million females; about one-third of all PUC, and less common in females than males.**
- **Two-thirds are over 60; incidence higher in Black females and may be falling.** European registries confirm **rarity, advanced age at presentation and male predominance**.
- Risk factors: **leukoplakia, glandular metaplasia, chronic irritation, stricture disease, recurrent urinary infection, caruncle, human papillomavirus infection.**
- **Urethral diverticula predispose — 6–10% harbour a malignancy.**

- **Nearly all are symptomatic: haematuria or vaginal spotting, lower urinary tract symptoms, vaginal or urethral mass.**

Anatomy and Pathology

COMPARISON: MALE VERSUS FEMALE URETHRA

Feature	Male	Female
Length	~20 cm	~4 cm
Segments	Anterior = bulbar, pendulous, fossa navicularis; posterior = prostatic, membranous	Anterior = distal one-third; posterior = proximal two-thirds
Epithelium	Posterior urothelium; anterior columnar; fossa navicularis squamous	Proximal third urothelium; distal two-thirds non-keratinising stratified squamous
Glands	Cowper's (bulbourethral), glands of Littre	Skene's (paraurethral) — female homologue of the prostate, draining near the meatus
Drainage	Anterior → inguinal nodes; posterior → pelvic nodes. Identical in both sexes	
Histology mix	Urothelial 50–80%, SCC 10–30%, adenocarcinoma 5–10%	Urothelial 28–45%, SCC 19–29%, adenocarcinoma 28–38% — near-equal
Continence	Distal urethrectomy does not threaten continence	The distal third can be excised without compromising continence — the fact that governs surgery

- Purpose: **the anterior–posterior division means something different in each sex** — in the male a change of epithelium, in the female also the continence boundary.
- Pattern: **lymphatic drainage is the one property that does not differ**, which is why inguinal doctrine is shared. Trap: importing male histological proportions — **adenocarcinoma is far more prominent in females.**
- **Adenocarcinoma is commoner in females, may be particularly prevalent in Black females, and predominates in urethral diverticula.** Two subtypes:
 - Mucinous — commoner; resembles colonic or cervical adenocarcinoma; may produce carcinoembryonic antigen (CEA).
 - Clear cell — resembles clear cell carcinoma of the female genital tract; can secrete prostate-specific antigen (PSA); commoner in diverticula.
- Some resemble prostatic adenocarcinoma microscopically. Likely origin the Skene's gland — possibly mucinous from the proximal gland, clear cell from the distal; alternatives are glandular metaplasia, urothelium, nephrogenic adenoma, residual Müllerian tissue.
- Rare histologies: melanoma, sarcoma, lymphoma, neuroendocrine, small cell carcinoma.

Evaluation and Staging

- Bimanual and pelvic examination with attention to vaginal wall and vulval involvement; assess groins for adenopathy.
- Cystourethroscopy and examination under anaesthesia with biopsy; percutaneous or transvaginal biopsy are alternatives.
- MRI gives the best anatomical detail of pelvis and urethra and assesses local extension and nodes; chest radiograph or CT for distant disease; bone scan only on clinical suspicion.
- Tumour location is critical to treatment, deciding whether continence-sparing excision is possible. One-third present distal; the remainder proximal or pan-urethral.

- **Serum PSA may be raised in a few females with adenocarcinoma and may normalise after successful treatment.**
- **Same TNM system as males.** At presentation: 30–50% locally advanced; ~20% clinically enlarged inguinal nodes, usually malignant; 10–20% pelvic nodal or distant metastases.
- **Females present at higher stage and grade than males; SCC carries particular risk of synchronous or metachronous nodal involvement.**

Prognosis

- **Aggressive disease — five-year survival 40–65% — but females have no independently increased risk of death versus males;** the worse crude outcome reflects **later stage at presentation.**
- **Stage is the dominant prognostic factor:** the earliest report held that survival is determined chiefly by extent of neoplasm, and all later work confirms it. Independently adverse: **higher stage, positive nodes, pan-urethral location, Black race, increased age, non-squamous histology.**
- **Location gradient distal > proximal > pan-urethral,** about a threefold spread between distal and pan-urethral; but **proximal and pan-urethral tumours are more often higher stage.**
- **Size gradient at <2 cm, 2–4 cm, >4 cm,** survival falling steeply — though **size correlates with location and stage and the analysis did not adjust for stage.**
- Unifying statement: **stage correlates with grade, size, location and nodal status, and stage drives prognosis.**
- **Histological subtype is contested:** several series find no difference and advise identical treatment regardless of histology; two large population studies found **SCC has a lower risk of death than non-squamous histologies;** one cohort ranked **SCC ≈ urothelial > adenocarcinoma.** **Urethral melanoma has a uniformly poor prognosis.**
- **Many historical series predate cross-sectional imaging and are incompletely staged — a caveat on every figure above.**

Treatment

- **Evidence is small institutional cohorts and case reports; there is no standardised algorithm.** Options: surgery, radiotherapy, chemotherapy, alone or combined. About 40% receive radiotherapy in some form.
- **Governing principle: local excision may suffice for small low-stage anterior tumours; proximal and advanced tumours require an aggressive approach.**

Anterior Tumours — Surgery

- **Anterior tumours tend to be low stage, and at least 70% are cured by local excision alone.**
- **Small non-invasive tumours — endoscopic resection or laser fulguration ± adjuvant BCG; if not endoscopically amenable — surgical resection or radiotherapy.**
- **Small exophytic anterior tumours excised with a portion of anterior vaginal wall transvaginally, with frozen section of the proximal urethra to confirm a negative margin.**
- **Outcomes: partial urethrectomy or endoscopic resection gives good five-year disease-specific survival, but about one in five recur locally and many cannot be salvaged.**
- **Complications: new or worsening stress urinary incontinence in over 40% despite the anatomical teaching; meatal stenosis, minimised by wide spatulation.**
- **Bladder-sparing options in selected advanced anterior tumours: radical urethrectomy with bladder neck closure and a catheterisable stoma, or ileovesicostomy, if a negative margin is achievable.**

- **Radical urethrectomy extent:** all periurethral tissue — laterally to the bulbocavernosus muscles, anteriorly to the pubic symphysis, superiorly to the bladder neck, including the anterior vaginal wall.

Anterior Tumours — Radiotherapy

- **Effective for low-stage anterior tumours**, and the alternative when urethral preservation is desired or surgery would compromise function.
- **External beam, brachytherapy or both**, at 55–70 Gy; distal tumours commonly brachytherapy alone at a median 60 Gy, with good five-year survival when only part of the urethra is involved.
- **Extent of urethral involvement was the only variable predicting local control** — length involved, not histology or size.
- **Complications 20–50%:** urethral stricture, necrosis, fistula, cystitis. Higher cumulative dose raises risk; contemporary technique lowers it.

Inguinal Lymph Nodes

- **Up to a third present with or develop inguinal adenopathy;** up to 20% recur in inguinal nodes, usually within two years.
- **As in males, no clear role for prophylactic inguinal lymphadenectomy in cN0 high-risk anterior tumours**, and unlike penile and vulval cancer, no prognostic factors predict inguinal metastasis. Precedent exists in vulval SCC, and further study has been advocated.
- **Prophylactic groin radiotherapy should be considered where the primary is irradiated;** it appears to reduce inguinal recurrence, though recurrences still occur despite it.
- **Palpable inguinal nodes require aggressive treatment** — usually metastatic, and some patients are curable. Guideline options: chemoradiotherapy, chemotherapy alone, inguinal lymphadenectomy, or chemoradiotherapy followed by consolidative surgery. Lymphadenectomy technique is identical to that for penile cancer.
- **Pelvic adenopathy carries a poor prognosis and patients usually die of disease;** benefit of pelvic lymphadenectomy for clinically positive pelvic nodes is unclear.

Posterior and Pan-urethral Tumours

- **More likely high stage, involving the whole urethra and invading bladder, vagina and external genitalia;** five-year metastasis-free survival only 24–48% despite aggressive surgery or radiotherapy.
- **Usually requires anterior pelvic exenteration**, which alone gives low cure rates and significant local recurrence — the majority treated by radical surgery alone died of disease and about a third recurred in the pelvis. Pelvic recurrence usually proves fatal.
- **Components:** radical cystectomy + urethrectomy + extended pelvic lymphadenectomy + wide or complete vaginal excision where margins demand; hysterectomy and bilateral salpingo-oophorectomy case by case; partial vulvectomy or labial excision if the lesion reaches the external genitalia.

Perineal portion of anterior exenteration.

1. **Inverted U-shaped incision widely encircling the urethral meatus.**
 2. **Extend onto the posterior vaginal wall and labia minora**, continuing anteriorly to include the clitoris where required.
 3. **En bloc resection of pubic symphysis and inferior pubic rami** if a negative pubic margin is otherwise unobtainable — intraoperative radiotherapy is an alternative to bone resection.
- Otherwise as for bladder cancer, with a more extensive perineal component to secure wide urethral margins.

- Radiotherapy for proximal tumours: **similarly poor local control, median survival under three years; combined external beam and brachytherapy outperform external beam alone, probably by dose escalation; up to 50% local failure, with length of urethra involved significantly affecting local control; almost all failures within two years and salvage may not be possible.**

Multimodal Therapy and Systemic Chemotherapy

- Rationale: **surgery and radiotherapy have similarly low local control and poor survival when used alone.**
- **At least 20% receive combined surgery and radiotherapy, and this is rising; multimodal therapy is associated with longer overall survival, though not significantly so in the institutional series examining it. Neoadjuvant radiotherapy then surgery may give better local control than either alone.**
- **Academic centres deliver more multimodal therapy and radical surgery; referral of advanced disease is recommended.**
- **Guidelines recommend multimodal therapy for advanced-stage female PUC — systemic chemotherapy with consolidative radiotherapy or surgery, or radiotherapy followed by surgery, with chemotherapy dictated by histology as tabulated above.**
- **Systemic chemotherapy for metastatic disease; commonest distant sites lung, bone and brain.**
- **Evidence is poor and confounded by treatment indication, so treatment is individualised — but multimodal treatment should be strongly considered in advanced disease.**

KEY POINTS: FEMALE URETHRAL CANCER

- **The three commonest histologies are urothelial carcinoma, SCC and adenocarcinoma.**
- **Compared with anterior urethral cancers, posterior cancers present at a more advanced stage and are associated with worse survival.**
- **Radiation and surgical excision are both options for low-stage anterior tumours, each with high cure rates.**
- **Surgery and radiotherapy for proximal female urethral tumours have poor outcomes when used alone; multimodal therapy is recommended.**

Concept — *everything is decided by one anatomical line: the distal third can be removed and the proximal two-thirds cannot, so distal disease has two curative options of equal standing and proximal disease has no adequate single modality at all.*

URETHRAL RECURRENCE AFTER RADICAL CYSTECTOMY

The Male Urethra

- **After RC the remnant urothelium of urethra and upper tracts remains at risk, by direct growth along the urothelium from the primary or metachronous development from a field defect.**
- **Early practice was prophylactic urethrectomy, abandoned because 64–100% of prophylactically removed urethras showed no disease, and where present it was most commonly CIS; even identified at-risk patients mostly had negative pathology.**
- **Three forces drove the shift to urethral preservation: low prevalence of unrecognised synchronous involvement, added morbidity of urethrectomy, growing use of orthotopic diversion.**
- **Risk of urethral recurrence (UR) after RC is 1–15%, larger series near 5%. Timing: most within two years; about 11% after five years; rarely as late as fifteen.**

- **Risk factors:** bladder tumour multifocality, CIS, history of NMIBC, prostatic involvement with urothelial carcinoma, positive urethral margin, cutaneous diversion. Risk rises with the number present, but applying all of them would exclude as many as 70% from orthotopic neobladder (ONB) eligibility — so only the highest-risk patients undergo prophylactic urethrectomy.

Prostatic Involvement, Margin and Diversion

- Prostate involved in 17–40% of RC specimens, most often prostatic urethra or ducts, sometimes stroma. Predictors: recurrent bladder tumours, trigone or bladder neck tumours, multifocality.
- Prostatic urethral involvement raises UR risk roughly threefold — 15% versus 5%; some groups advise against ONB where present.
- Preoperative transurethral prostatic biopsy has good sensitivity and high negative predictive value for prostatic involvement, but correlates poorly with the final urethral margin — the diagnosis it makes is not the question asked — so it has largely been abandoned in favour of intraoperative frozen section.
- A positive urethral margin occurs in up to 7% of RC specimens and is a strong UR risk factor, in one cohort the strongest after controlling for prostatic involvement and stage. Many URs occur without a positive apical margin and some groups found no significant association, so other factors contribute.
- Most surgeons proceed with ONB if frozen section of the distal prostatic urethral margin is normal; frozen section is accurate and a positive result can often be converted with further resection.
- Dissenting view: most patients with a positive margin never develop a UR, and frozen section itself risks urethral shortening and false positives; one ONB cohort used none, with a low UR rate and no death from UR.
- The American Urological Association Muscle-Invasive Bladder Cancer guideline recommends verifying a negative urethral margin before ONB, and concern for invasive carcinoma at the margin mandates urethrectomy rather than ONB.
- ONB carries lower UR risk than cutaneous diversion — about 5% versus 9% at five years, a halved relative risk. Beyond selection bias, proposed mechanisms are altered local immune response, antineoplastic effects of ileum, continued exposure of the urethra to urine.
- In the largest cohort, after adjustment for stage and margin, significant associations were ONB (protective), CIS, prostatic urethral involvement, bladder tumour multifocality.

Prophylactic Urethrectomy

- *Relative, advocated by some:* multifocal bladder tumours, upper tract tumours, diffuse CIS, prostatic urethral involvement, bladder neck involvement.
- *Absolute:* positive urethral margin or gross urethral involvement.
- Staged urethrectomy after RC has outcomes similar to planned urethrectomy at RC — timing is not critical.
- Total urethrectomy after RC with cutaneous diversion is challenging, particularly excising the proximal urethra from the pelvic floor; that segment is at risk of positive margins, and incomplete resection causes retropubic relapse that is difficult to manage.

Presentation, Surveillance and Treatment

- URs present with bloody urethral discharge or a palpable mass, tending to be proximal but occurring anywhere.
- Cytology: voided sample with an ONB, catheterised urethral barbotage with a cutaneous diversion. Screening cytology has not been shown to improve survival — no difference between symptomatic and cytologically detected recurrences, though symptomatic patients more often had invasive disease

at urethrectomy. Guidelines still recommend considering urethral wash cytology for two years after RC.

- **Positive cytology or symptoms (bleeding, discharge, pain, palpable mass) require urethroscopy and biopsy; false positives occur, so confirm before treating. Pelvic CT or MRI and full restaging.**
- **Total urethrectomy is the treatment of choice, and the entire urethra must go** — incomplete removal of the distal urethra leaves a risk of distal recurrence.
- **Urethrectomy with a cutaneous diversion risks a positive proximal margin from the scarred pelvic floor. After orthotopic diversion, radical urethrectomy generally requires conversion to a cutaneous diversion, though subtotal urethrectomy with perineal urethrostomy is described.**
- **Non-invasive URs: endoscopic resection and topical therapy. Intraurethral BCG is cumbersome but has high response rates for urethral CIS — not for papillary or invasive disease. Significant risk of a second UR after endoscopic management. Radiotherapy has also been used.**
- **Prognosis is conflicting** — a large meta-analysis found no impact of UR on survival, a large single-institution cohort found reduced cancer-specific survival. The split that matters: **isolated non-invasive UR has a favourable prognosis and is usually salvageable; invasive UR is aggressive with median survival 17 months. At least one-third have concomitant pelvic or distant recurrence, and then prognosis is poor.**

TABLE 83.5 MALE URETHRAL RECURRENCES AFTER RADICAL CYSTECTOMY

Cohort type	Urethral recurrence	Time to recurrence
Historical, all diversions (1960s–1980s)	4–15%	Mean 12–26 months
Contemporary, all diversions (1990s onward)	1–7%, clustering near 5%	Median 13–25 months
Orthotopic neobladder only	1–5%	Median 17 months to 2.3 years; one series 66 months
All series combined	1–15%, larger series near 5%	12–30 months ; range 6 months to 5 years

Source: compiled series of cystectomies performed between 1960 and 2018.

- **Purpose: UR is uncommon and becoming less common. Pattern: rates fall as series become contemporary and orthotopic diversion predominates** — simultaneously a selection effect and a real protective effect.
- **Trap: reading the low contemporary rate as permission to stop surveillance at two years. The tail extends past five years.**

Total Urethrectomy After Cutaneous Diversion

1. **High lithotomy, hips and knees gently flexed, limbs abducted in stirrups; urethral catheter aids dissection.**
2. **Modified lambda-shaped or midline perineal incision; divide and retract subcutaneous tissue and bulbospongiosus in the midline to expose the corpus spongiosum.**
3. **Mobilise the spongiosum circumferentially at the bulbar urethra; traction allows sharp distal dissection separating spongiosum from corpora cavernosa.**
4. **Distally, the penis inverts, the corpora bow, the glans recedes into the phallus — the penis is turned inside out onto the perineum; continue to the base of the glans.**
5. **Replace the penis anatomically to excise meatus and glandular urethra; incise around the meatus, down each side of the ventral glans.**

6. **Free the distal urethra from its investments within the glans; deliver the pendulous urethra onto the perineum.**
 7. **Reapproximate the deep spongiosum of the glans with 4-0 polydioxanone; close the surface with interrupted 4-0 chromic.**
 8. **Sharp proximal dissection of the bulb posteriorly and laterally, close to the bulb but avoiding entry.**
Hazard: bulbar arteries lie posterolaterally — control with electrocautery or suture ligature.
 9. **Detach the urethra from the corpora anteriorly to where it leaves the bulb, leaving the specimen attached only by the membranous urethra.** Hazard: **bowel adherent to the superior surface of the urogenital diaphragm in cutaneous diversions — dissect under direct vision, aided by separating the crura in the midline to open the intracrusal space.**
 10. **Excise completely the ill-defined fibrotic band remaining of the membranous urethra; frozen section confirms a negative proximal margin.**
 11. **Suction drain in the urethral bed through the perineum; close bulbospongiosus, subcutaneous tissue and skin with interrupted absorbable sutures; light pressure dressing.**
- **Complications uncommon: superficial haematoma, penile shaft oedema, infection.**

Total Urethrectomy After Orthotopic Diversion

1. **Abdominoperineal approach, lithotomy with adjustable stirrups, catheter into the neobladder.**
2. **Urethrectomy as above, to the membranous urethra.**
3. **Abdominal exploration with lysis of adhesions and mobilisation of the ONB to the urethral anastomosis.**
4. **Working from above and below, free the urethroneovesical anastomosis; excise a circular area of pouch with the urethra for an adequate margin; deliver through the perineum.** Hazard: **bleeding from tunnel musculature after excision of the membranous urethra — suture ligature or electrocautery.**
5. **Diversion usually by ileal conduit, often from bowel of the ONB itself, reconfigured as needed — incise along visible lines of previous closure, preserve the mesentery, excise the remaining pouch.**
6. **An afferent limb, if present, can become the conduit without reimplanting the ureters.**
7. **Conversion to a continent cutaneous diversion may be possible, by anatomy and patient motivation.**

The Female Urethra

- **Urethrectomy at female RC is easy and was traditionally standard, but better understanding of the female continence mechanism and the low rate of urethral involvement have made preservation with ONB popular.**
- **Synchronous urethral involvement at female RC is 8–13%; the main risk factor is bladder neck involvement, present in up to 25% of female cystectomy specimens.**
- **The decisive asymmetry: nearly all urethral tumours have bladder neck involvement, but only 30–50% with bladder neck involvement have urethral tumours — so bladder neck involvement is not an absolute contraindication to preservation.**
- **Anterior vaginal wall involvement may also increase risk, though it usually coexists with bladder neck involvement. Direct extension is the likeliest mechanism, but skip lesions occur.**
- **Proposed ONB criteria: absence of bladder neck, urethral and anterior vaginal wall involvement plus a negative intraoperative frozen section, which most surgeons regard as essential and which can often be converted from positive to negative with further resection.**

- **Trigonal involvement or positive nodes do not preclude ONB provided the urethral frozen section is negative.** Females not having an ONB should undergo concurrent urethrectomy.
- **UR after ONB in females is rare — under 5%, explained by less at-risk urothelium in a shorter urethra and cephalad extension of squamous epithelium after the menopause.**
- **Presents with obstruction, haematuria and pain, most commonly in the proximal urethra, usually two to four years postoperatively, though one recent report found a median of eight months.**
- **In large selected ONB cohorts, UR rates were very low and only one of seven recurrences had a positive final margin at RC — most therefore arose from skip lesions.**
- **Treatment is usually urethrectomy with conversion to cutaneous diversion, though BCG is reported. Isolated UR has a good prognosis with curative treatment; UR with synchronous pelvic recurrence, distant metastasis or locally advanced disease is dismal.**

COMPARISON: MALE VERSUS FEMALE URETHRAL RECURRENCE AFTER RADICAL CYSTECTOMY

Feature	Male	Female
Incidence	1–15%, larger series near 5%	Under 5% after ONB; synchronous involvement at RC 8–13%
Principal risk factor	Prostatic urethral involvement; also positive margin, cutaneous diversion, CIS, multifocality	Bladder neck involvement; also positive urethral margin, anterior vaginal wall involvement
Site	Usually proximal , but anywhere	Usually proximal
Timing	Most within 2 years; 11% after 5 years; up to 15 years	Usually 2–4 years; one series median 8 months
Mechanism	Direct extension or field defect	Mostly skip lesions — recurrence despite negative final margins
Presentation	Bloody urethral discharge, palpable mass	Obstruction, haematuria, pain
ONB eligibility test	Negative intraoperative frozen section of the urethral margin — identical in both sexes	
Treatment	Total urethrectomy; endoscopic resection ± intraurethral BCG if non-invasive	Urethrectomy with conversion to cutaneous diversion; BCG reported

- **Purpose: the sexes share a management algorithm but not a mechanism. Pattern: male recurrence is predicted by what the pathologist finds; female recurrence often is not, because skip lesions escape margin assessment.**
- **Trap: assuming a negative final margin protects the female urethra — it does not, which is why surveillance continues regardless.**

KEY POINTS: URETHRAL RECURRENCE AFTER RADICAL CYSTECTOMY

- **Male urethral recurrence after radical cystectomy is rare but more common with prostatic involvement, a positive urethral margin, and cutaneous diversions.**
- **The primary risk factors for female urethral recurrences are bladder neck involvement and a positive urethral margin.**
- **Most surgeons will proceed with ONB if an intraoperative frozen section of the urethral margin is negative.**

- **After urethrectomy in patients with ONBs, bowel from the neobladder can often be reconfigured and used for a cutaneous diversion.**

Concept — *this is a risk-stratification problem rather than an oncological one: the event is rare enough that removing every at-risk urethra would deny most patients a neobladder for no benefit, so frozen section, risk factors and cytology exist only to identify the small minority in whom the urethra must go.*